

What is Agentic AI?

Agentic AI refers to artificial intelligence systems that can **act autonomously**, make decisions, and **perform tasks on behalf of a user or system**—often with minimal human intervention.

Unlike traditional AI (which only responds), Agentic AI can:

- **Plan**
 - **Reason**
 - **Take actions**
 - **Adapt based on outcomes**
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Simple Definition

 **Agentic AI = AI that behaves like an “agent”**

It doesn't just generate outputs — it **takes initiative to achieve a goal**.

How Agentic AI Works

1. **Goal is defined**
(e.g., “Fix a server issue”)
 2. **Planning**
 - Breaks the goal into steps
 3. **Execution**
 - Calls tools, APIs, scripts
 4. **Observation**
 - Checks results
 5. **Iteration**
 - Adjusts actions until goal is achieved
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Key Characteristics

-  Autonomous decision-making
-  Multi-step reasoning
-  Tool usage (APIs, scripts, databases)
-  Memory & context awareness
-  Feedback-driven improvement

Agentic AI vs Generative AI

Feature	Generative AI	Agentic AI
Role	Content generation	Task execution
Example	ChatGPT writing text	AI fixing production issues
Behavior	Reactive	Proactive
Output	Single response	Multi-step actions

Real-Time Use Cases (DevOps / AWS / Monitoring)

1. Incident Management

- Detects alert in monitoring tool
- Analyzes logs
- Restarts service automatically
- Notifies team

 Example tools: Dynatrace, Prometheus, CloudWatch

2. Cloud Automation (AWS)

- Identifies high CPU usage
 - Scales EC2 automatically
 - Optimizes cost
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3. 🔍 Root Cause Analysis

- Correlates logs + metrics
 - Identifies failing microservice
 - Suggests or executes fix
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4. 🧪 CI/CD Pipeline Optimization

- Detects pipeline failure
 - Reruns failed stage
 - Applies fixes (dependency/version issues)
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✂️ Examples of Agentic AI Systems

- AutoGPT
 - AI-powered DevOps bots
 - Autonomous customer support agents
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📊 Why Agentic AI Matters

- Reduces manual effort
 - Faster incident resolution
 - Improves system reliability
 - Enables self-healing systems
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📄 Summary

👉 Agentic AI = Goal-driven, action-oriented AI

It goes beyond “thinking” (Generative AI) and moves into “**doing**”.